

A 1000 μF capacitor is charged to a potential difference of 12 V.
How much energy is stored in the capacitor?

Solution:

Use:

$$E = \frac{1}{2}CV^2$$

Convert capacitance:

$$C = 1000 \mu\text{F} = 1000 \times 10^{-6} = 1.0 \times 10^{-3} \text{ F}$$

$$E = \frac{1}{2} \times 1.0 \times 10^{-3} \times 12^2 = 0.072 \text{ J}$$

Answer: 0.072 J

Revision Tip:

Always convert microfarads (μF) to farads before calculating. Use $E = \frac{1}{2}CV^2$ for stored energy.